

TODO — Outstanding & Deferred Work

Tracked against the wave roadmap (`configs/wave_registry.json`). Items below are the honest backlog after the v6.0 (W003+W004) build.

High priority (data fidelity)

- [] **W005 — XLSX tag/instrument register reconciliation + coverage delta** (NEXT, top priority).
Reconcile the 297 auto-catalogued components against the official tag/instrument register (XLSX): report which are matched, which are **false positives**, which register tags are **missing**, plus nomenclature reconciliation (our segment labels vs official naming). Cross-validate colour-derived line identity against ISA tag class (`configs/isa_classes.json`). This is the question an engineering reviewer asks first.
- [] **Recover the U line** (top-left). Currently rendered black; needs colour re-attribution so it joins the parent-header tree alongside A/B/D/E/W/S/V.
- [] **Map the 112 residual UNRESOLVED_OTHER_COLOUR elements** (magenta / non-canonical family). Requires augmenting the legend swatch table or a manual engineering call.
- [] **Bind the 77 floating arrows** to their source lines (nearest-line / direction heuristic) so flow topology is complete.

Medium priority (layer completeness)

- [] Populate **04G_E_RED** line-geometry layer (RED is currently text-only).
- [] Populate **03A_Manifold_COLD_Header** and **03B_Manifold_WARM_Header** layers.
- [] Verify handover-diamond category prefix per **AD_01.16** (continuity flag #2).
- [] Visual review of the **W-line bottom-right zone** (19 elements, atlas layer 04C).

Future waves (planned, not started)

- [] **W005** — Tag & Instrument Register Reconciliation (XLSX coverage delta). (active, HIGH — see top)
- [] **W006** — Interactive layered viewer (layer toggle / colour-line isolation / click-through).
- [] **W007** — CI (GitHub Actions) + cross-drawing reconciliation + temperature/pressure annotation.
- [] **W008** — Round-trip reassembly (recompose isolated lines into full P&ID).
- [] **W009** — Publication & engineering sign-off.

Engineering / infra (deferred)

- [] CI / GitHub Actions workflow (planned in `docs/GITHUB_PR_PLAN.md`).
- [] Web UI / interactive app beyond the current static HTML atlas.
- [] Expand automated test coverage (currently smoke + count assertions).